Time Series Agent: Components and Information Flow

§3 LLM SIDE

interprets temporal evidence and selects actions

- Representation alignment
- Temporal instruction & reasoning
- Workflow policy learning

ChatTS · Time-LLM · TimeRLM

§4 GENERAL HARNESS

executes actions, keeps state, coordinates roles

- Interface & Interaction
- Memory & Context
- Tools & Execution
- Planning & Orchestration
- Optimization & Evolution

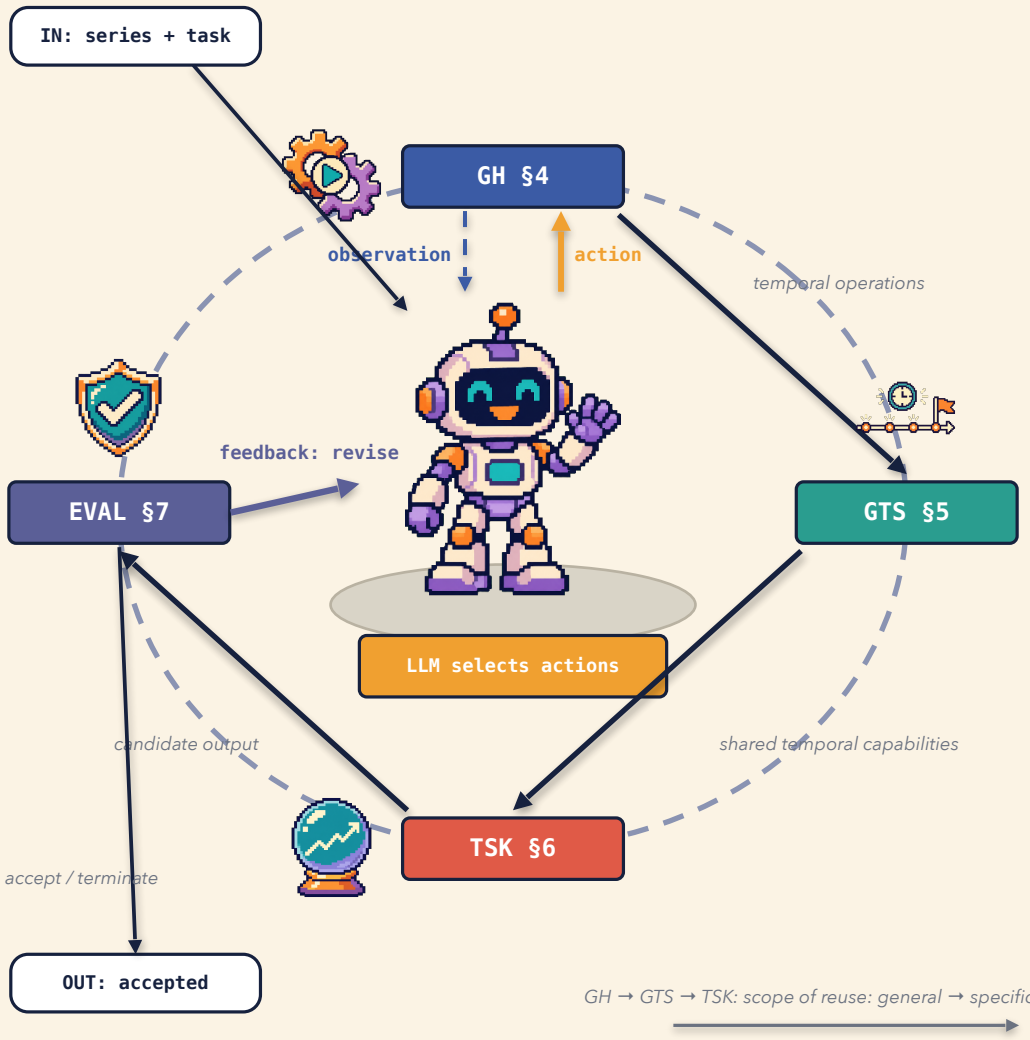
TimeClaw-Harness · AION · TS-Agent · TradingAgents

§5 GENERAL TIME SERIES HARNESS

supplies temporal data, tools and time constraints

- Temporal Data Management
- Temporal Processing
- Temporal Context & Memory
- Temporal Validation

TimeSage-EV · OpenForecaster · TSQAgent · TS-Reasoner



§6 TASK-SPECIFIC HARNESS

organizes outputs around task objectives

- Forecasting & Prediction
- Data Augmentation & Synthesis
- Anomaly Detection & Diagnosis
- Decision Support

FLAIRR-TS · MetaCaster · Argos · FinCon

§7 VALIDATION & EVALUATION

checks process, temporal validity, task and decision utility

- Execution checks
- Information-cutoff checks
- Task metrics
- Downstream utility

TimeSeriesGym · Dr-CiK · TraceBench · TSFLab

IN Time series & context
observations · events · task request

OUT Forecast · diagnosis · decision
accepted output